Boosting Faithful Multi-Modal LLMs via Complementary Visual Grounding

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***Abstract*—Multimodal Large Language Models (MLLMs) exhibit impressive performance across vision-language tasks, but still face the hallucination challenges, where generated texts are factually inconsistent with visual input. Existing mitigation methods focus on surface symptoms of hallucination and heavily rely on post-hoc corrections, extensive data curation, or costly inference schemes. In this work, we identify two key factors of MLLM hallucination: Insufficient Visual Context, where ambiguous visual contexts lead to language speculation, and Progressive Textual Drift, where model attention strays from visual inputs in longer responses. To address these problems, we propose a novel Complementary Visual Grounding (CVG) framework. CVG exploits the intrinsic architecture of MLLMs, without requiring any external tools, models, or additional data. CVG first disentangles visual context into two complementary branches based on query relevance, then maintains steadfast visual grounding during the auto-regressive generation. Finally, it contrasts the output distributions of two branches to produce a faithful response. Extensive experiments on various hallucination and general benchmarks demonstrate that CVG achieves state-of-the-art performances across MLLM architectures and scales.**

***Index Terms*—Multimodal learning, vision-language models, hallucination mitigation, model faithfulness.**

I. Introduction

RECENT breakthroughs in Large Language Models (LLMs) have spurred the development of highly capable multi-modal LLMs (MLLMs) [1], [2]. By processing visual modalities in tandem with text, MLLMs enable a wide array of vision-language downstream tasks, including cross-modal retrieval [3], [4], [5], image/video captioning [6], [7], [8], [9], and visual question answering [10], [11], [12]. MLLMs have captured the focus of researchers for their profound ability to comprehend the visual world and engage in conversational interactions with users [13], [14]. MLLMs typically extend the LLMs by incorporating a vision encoder. This encoder transforms visual inputs into patch embeddings, which are then projected into the textual embedding space of the language model. This cross-modal alignment enables the LLM to jointly process visual and textual information. Despite achieving impressive performance across various multi-modal tasks, existing MLLMs are prone to hallucinations [15], where the model generates text that is factually inconsistent with the provided visual content, as shown in Figure 1(a). Such hallucinations critically undermine the faithfulness of MLLMs [16], posing a significant barrier to their deployment in real-world applications.

Various approaches have been proposed to mitigate hallucinations in MLLMs. One straightforward approach involves post-hoc correction, which employs external tools to revise hallucinations [17], [18], [19]. They first use advanced models to detect hallucinations for original responses, and then use specific knowledge to revise hallucinated contents. These methods rely on the strong ability of external models to construct elaborate workflows. Another approach is to enhance supervised fine-tuning with diverse instruction data [20] or reinforcement learning from human/AI feedback [21], [22]. These methods require extensive, high-quality annotated data, making them labor-intensive and costly to scale. The training-free approach explores multiple decoding paths to rectify hallucination [23], [24], [25], bringing extra inference cost. While the above approaches offer valuable strategies, they just focus on the surface symptoms of MLLM hallucination. In this paper, we identify two key factors that inherently induce hallucination, as shown in Figure 1(b) and (c).

The first factor is the **Insufficient Visual Context**: MLLMs are prone to hallucination when a query requires information from ambiguous or incomplete visual content. This inadequacy compels MLLMs to over-rely on underlying language knowledge instead of grounding in visual information. To validate this, our experiments (Figure 2) reveal that MLLMs dynamically focus their visual attention on query-relevant regions. Consequently, when these specific regions are masked to simulate insufficient context, the incidence of hallucination increases significantly (higher than random masking). This finding demonstrates that visual completeness is critical; an inadequate visual context forces the model to default to its linguistic priors, thereby inducing hallucinations.

The second factor is **Progressive Textual Drift**. During the generation of long responses, the model's attention to the visual context can wane over time. This decay leads the model to drift from visual grounding toward its internal language patterns, resulting in hallucinations. As validated by our experiments in Figure 3, visual attention is strongest for initial tokens but progressively diverges and weakens as the response

Received 3 August 2025; revised 20 November 2025; accepted 9 December 2025. Date of publication 22 December 2025; date of current version 29 December 2025. This work was supported in part by the Artificial Intelligence-National Science and Technology Major Project under Grant 2023ZD0121200 and in part by the National Natural Science Foundation of China under Grant 62222212 and Grant 62121002. The associate editor coordinating the review of this article and approving it for publication was Dr. Huafeng Li. *(Corresponding author: Zhendong Mao.)*

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Digital Object Identifier 10.1109/TIP.2025.3644140

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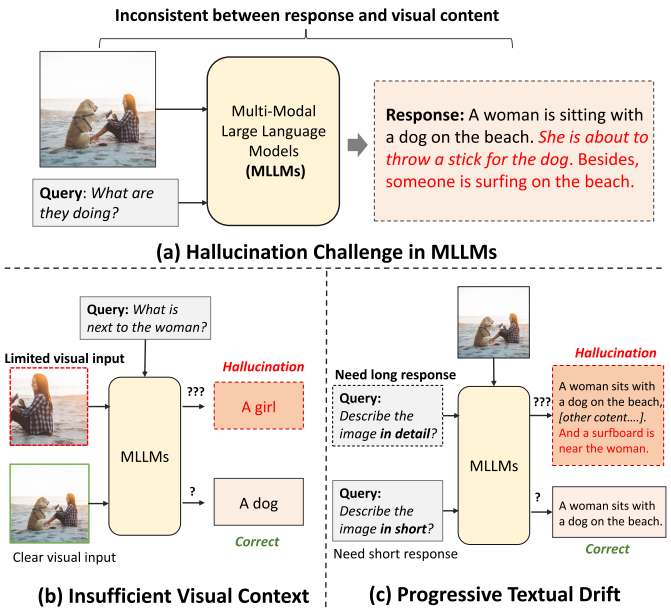


Fig. 1. **An overview of the hallucination challenge in Multi-modal Large Language Models (MLLMs) and the two key factors investigated in this paper**. (a) Hallucination Example: The model generates a response that is factually inconsistent with the visual input. (b) Insufficient Visual Context: Hallucinations are more probable when visual information is limited or incomplete to the user's query, compelling the model to guess based on its internal knowledge. (c) Progressive Textual Drift: Hallucinations are more probable in longer responses, since the model's attention gradually shifts away from visual grounding and towards its own language patterns.

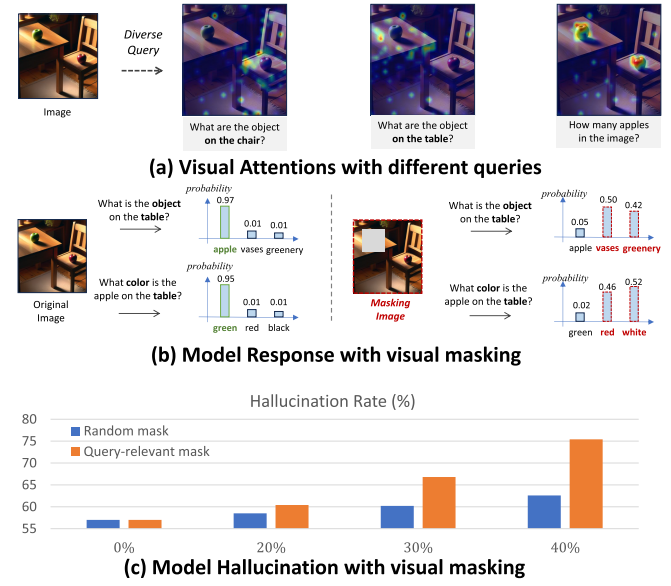


Fig. 2. **The experimental support for the insufficient visual context**. (a) MLLMs exhibit varied visual attention patterns for different queries on the same image, consistently focusing on query-relevant regions. (b) The model generates correct answers with complete visual context. However, masking key visual regions leads to insufficient context, causing erroneous hallucinations. (c) Quantitative analysis on the CHAIR benchmark [26] shows that masking increases hallucination rates, with a more pronounced effect when query-relevant regions are obscured compared to random masking.

lengthens. This decay in visual grounding directly correlates with higher hallucination rates in longer outputs. This trend indicates that as MLLMs generate outputs, their reliance shifts

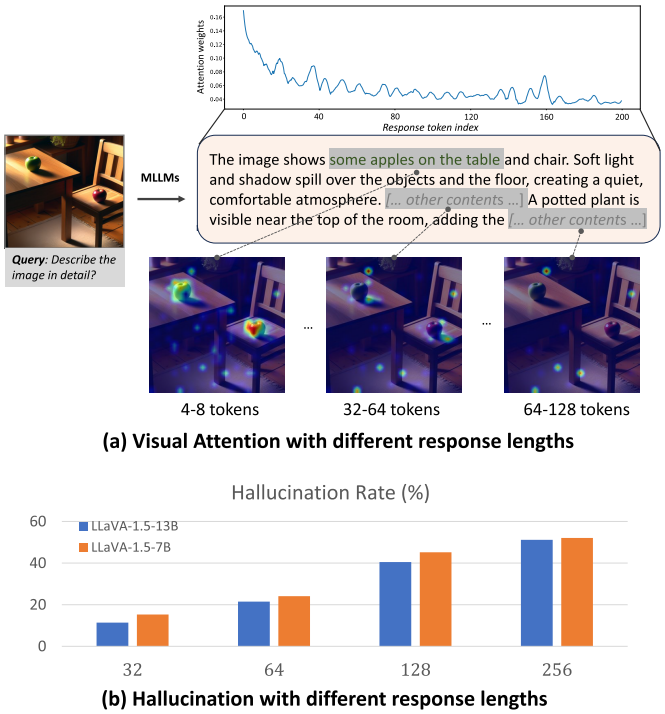


Fig. 3. **The experimental support for the progressive textual drift**. (a) Evolution of visual attention during generation. Initial visual attention correctly focuses on relevant regions but diverges and weakens as responses lengthen. (b) Quantitative analysis on the CHAIR benchmark [26] reveals that MLLMs exhibit significantly higher hallucinations with longer responses.

from factual visual grounding to inherent linguistic patterns, thereby increasing the propensity for hallucination.

To address these challenges, we introduce a novel **Complementary Visual Grounding (CVG)** framework to handle both insufficient visual context and progressive textual drift challenges. Unlike previous works, CVG requires no post-hoc corrections, extensive data curation, or external models, just leverages the intrinsic architecture of MLLMs to mitigate hallucinations. As shown in Figure 4, the framework comprises three key modules: First, the Visual Context Disentanglement module identifies query-relevant visual regions using an attention mechanism. It then bifurcates the visual tokens into two complementary sets: a chosen set of query-relevant tokens and a rejected set of trivial ones. These two sets establish parallel evidence branches for the MLLM. Second, the Dynamic Attention Calibration module exercises adaptive visual control during auto-regressive generation. It applies a visual-anchored calibration to the attention scores for both evidence branches, ensuring that subsequent generation remains grounded in visual inputs and suppressing the drift tendency. Finally, the Grounding Flow Fusion module employs a contrastive fusion on the output distributions from two complementary branches and produces a robust decoding output. Therefore, CVG could ensure comprehensive visual grounding to mitigate hallucinations and enhance the faithfulness of MLLMs.

We conducted extensive experiments on various hallucination benchmarks, including both discriminative and generative tasks. Besides, we evaluated CVG on a wide range of general-purpose benchmarks. The experimental results confirm the superiority of our framework in mitigating hallucinations while

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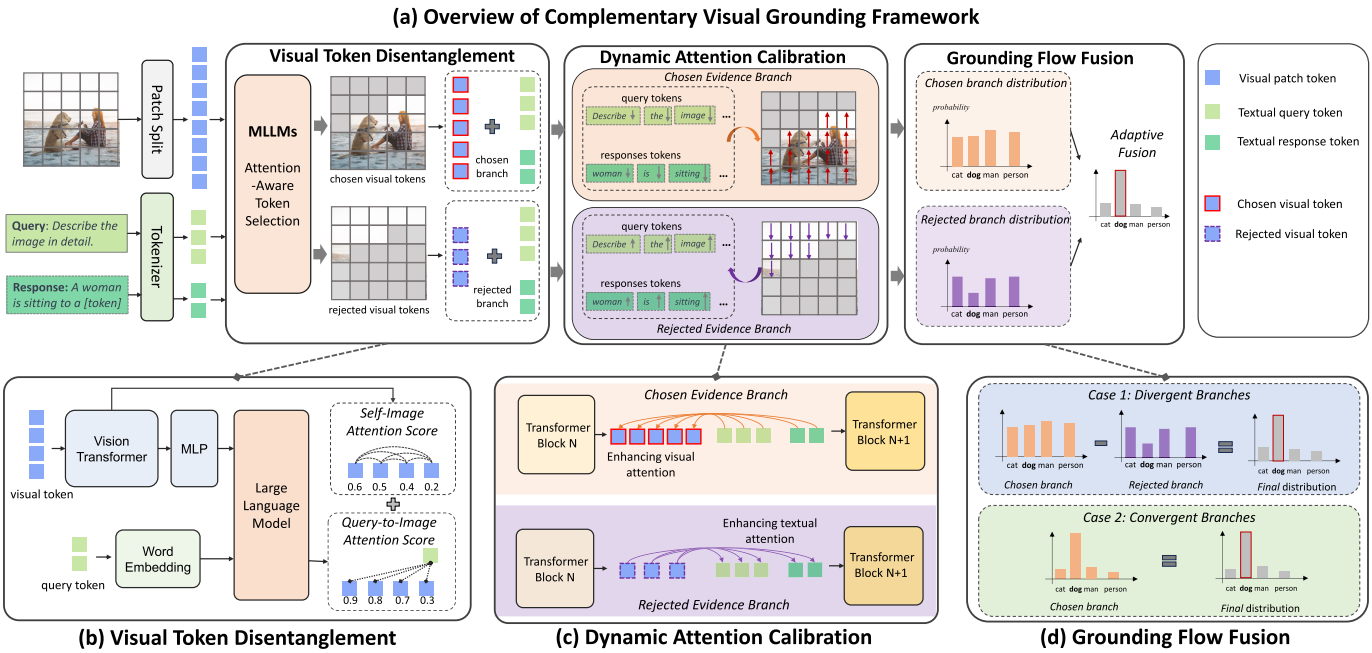


Fig. 4. (a) **Overview of our Complementary Visual Grounding (CVG) framework**. The framework first disentangles input visual tokens into two complementary sets: a chosen set with query-relevant tokens and a rejected set with trivial ones. These sets form two evidence branches for MLLMs. Then it performs a visual-anchored calibration of attention weights across both branches. Finally, the output distributions from both branches are integrated by the contrastive decoding to produce the well-grounded response. (a) **Visual Token Disentanglement** module categorizes visual tokens into chosen and rejected groups, which relies on combining self-attention from the vision transformer and query-to-image attention from the LLM backbone. (c) **Dynamic Attention Calibration** module dynamically increases the attention for visual tokens in the chosen branch, and decreases those in the rejected branch, to ensure the distinction of visual grounding during the generation process. (d) **Grounding Flow Fusion** module contrasts the distributions from the chosen and rejected branches, which steers the generation towards tokens supported by the chosen evidence and away from those suggested by the rejected evidence.

enhancing overall performance, consistently across different base models and parameter scales. The contributions of this work are summarized as follows:

- We systematically identify and empirically validate two key factors of MLLM hallucination: Insufficient Visual Context and Progressive Textual Drift. This analysis offers a new perspective and shifts the focus from symptom treatment to addressing core mechanisms.
- We propose an effective and model-agnostic Complementary Visual Grounding (CVG) framework. By establishing and integrating complementary evidence branches, CVG ensures model reasoning remains grounded in a comprehensive understanding of the visual context.
- We design three core modules for the CVG framework: Visual Context Disentanglement, Dynamic Attention Calibration, and Grounding Flow Fusion. These modules work in concert to identify, maintain, and fuse visual evidence, preventing hallucinations during generation.
- We performed extensive experiments demonstrating that CVG achieves state-of-the-art performance on multiple hallucination and general-purpose benchmarks. The results validate the effectiveness and generalizability.

II. Related Work

A. Multi-Modal Large Language Models

The success of pre-training and instruction-tuning in Large Language Models (LLMs) has catalyzed research into MLLMs, which are designed to comprehend visual signals. These models aim to unify visual perception and language reasoning within a single framework. Early architectural approaches focused on this integration, either by using cross-attention mechanisms to compress visual information into query embeddings [1], [27] or by employing MLP projection layers to map visual features into the LLM's embedding space [2]. A significant advancement came with the introduction of visual instruction tuning [28]. By fine-tuning models on diverse image-text instruction pairs, this method greatly enhanced their interactive capabilities, enabling them to respond to visual queries in natural language. Subsequently, the development of larger and higher-quality multi-modal datasets has further advanced the field, pushing MLLMs to achieve superior visual understanding by training on more complex scenes and challenging reasoning tasks [29].

Despite this rapid progress, MLLMs are fundamentally adaptations of pre-trained LLMs and thus inherit their core characteristics. A critical consequence is that MLLMs exhibit a strong linguistic prior, instilled by the massive, text-centric corpora used to train their foundational models. This inherent bias can cause the model's learned linguistic patterns to overshadow the visual input. Consequently, the model may fail to properly ground its responses in the provided visual evidence, leading to inaccurate inferences and a flawed understanding of the multimodal context [14].

B. Hallucination Challenges in MLLMs

MLLMs have a concerning challenge that exhibits a pronounced tendency to generate hallucination responses [16]. The responses appear plausible but are factually spurious

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when analyzed against the provided visual content. To address this, various mitigation strategies have been proposed in the literature. One direct approach involves post-hoc correction [17], [18], [19], where hallucinations are mitigated after the initial response has been generated. These techniques typically employ external modules to verify and edit model outputs. For instance, they train hallucination detectors on supplementary data; once a hallucination is identified, the content is regenerated or corrected by a specialized revisor model. Although these methods can improve faithfulness, they add complexity and latency to the inference pipeline, increasing deployment costs. Alternatively, other methods aim to address hallucination during the model's training phase, primarily through supervised fine-tuning [20] or reinforcement learning with human/AI feedbacks [21], [22]. While often effective at strengthening cross-modal alignment, these methods require extensive fine-tuning or even architectural modifications, rendering them computationally expensive and time-consuming. A third category of methods focuses on the training-free decoding process [23], [24], [25]. These methods leverage multiple decoding paths or attention adjustments to revise the output distributions, without any additional model training.

C. Visual Grounding

Visual grounding aims to localize image regions corresponding to specific textual queries. It is an important downstream task for multi-modal understanding. Early work improves grounding by learning landmark-aware visual features [30], while subsequent methods enhance robustness through hierarchical phrase decoupling [31]. More recent studies introduce dynamic multi-modal decoders to model visual–textual interactions [32], and leverage dual-task frameworks to strengthen temporal–spatial alignment [33]. Although our framework does not directly address visual grounding, it leverages the underlying principle of precise cross-modal correspondence that characterizes these methods.

III. Preliminaries

Multimodal Large Language Models (MLLMs) are typically composed of three core components: a vision encoder, a projector, and a large language model (LLM) that serves as the language [14]. The vision encoder aims to extract the token sequence of visual embeddings from the input image. Then, the projector aligns visual embeddings with the textual feature space. This alignment enables the visual tokens to be concatenated with the textual token of a user's query, forming a unified input sequence. Finally, the pre-trained LLM, acting as the language decoder, processes this combined sequence to generate a textual response autoregressively.

A. Image Encoder

Given an input image I, a Vision Transformer (ViT) typically serves as the vision encoder. The image I is partitioned into a grid of N non-overlapping patches, and becomes a sequence of visual tokens with d-dimension:

$$\boldsymbol{X}_v = \mathrm{ViT}(I) = \{\boldsymbol{v}_1, \cdots, \boldsymbol{v}_N\} \in \mathbb{R}^{N\times d}, \tag{1}$$

B. Projector

Following the vision encoder, MLLMs employ a projection module, often a Multi-Layer Perceptron (MLP), to map the visual embeddings $\boldsymbol{X}_v$ into the LLM's input embedding space. This transformation is applied to each visual embedding independently, thus preserving the sequence length N. For notational simplicity, we continue to denote the sequence of projected visual embeddings as $\boldsymbol{X}_v$.

C. Language Decoder

MLLMs adopt pre-trained LLMs as the language decoder to leverage their powerful text generation capabilities [16]. A textual query Q in natural language is tokenized into a sequence of M textual embeddings:

$$\boldsymbol{X}_q = \mathrm{Tokenizer}(Q) = \{\boldsymbol{t}_1, \cdots, \boldsymbol{t}_M\} \in \mathbb{R}^{M\times d}, \tag{2}$$

Subsequently, the visual tokens $\boldsymbol{X}_v$ are concatenated with the query tokens $\boldsymbol{X}_q$. This combined sequence is then fed into the language decoder. Central to the decoder is the self-attention mechanism. Within each Transformer layer, an attention head h computes its output as:

$$\boldsymbol{O}_h = \boldsymbol{A}_h \boldsymbol{V}_h, \quad \boldsymbol{A}_h = \mathrm{softmax}\left(\boldsymbol{Q}_h \boldsymbol{K}_h^\top / \sqrt{d_k}\right), \tag{3}$$

Each head operates on its own query $\boldsymbol{Q}_h$, key $\boldsymbol{K}_h$, and value $\boldsymbol{V}_h$ matrices, all in $\mathbb{R}^{n\times d_k}$, where n represents the sequence length and d_k represents the hidden dimensions. The output O_h is a weighted sum of the value vectors, with the attention weights $\boldsymbol{A}_h \in \mathbb{R}^{n\times n}$ determining the focus of each token. This mechanism allows the model to adaptively integrate information from different parts of visual and textual input.

The response is generated token-by-token. The probability of generating the next token $(\boldsymbol{X}_r)_{k+1}$ is conditioned on the visual embeddings $\boldsymbol{X}_v$, the query embeddings $\boldsymbol{X}_q$, and the previously generated tokens $(\boldsymbol{X}_r)_{1:k}$, can be formatted as:

$$(\boldsymbol{X}_r)_{k+1} = p_{\mathrm{model}}\left(\boldsymbol{y} \mid \boldsymbol{X}_v, \boldsymbol{X}_q, (\boldsymbol{X}_r)_{1:k}\right) \quad k \in [1, K-1], \tag{4}$$

where $\boldsymbol{y}$ is the output tokens of MLLM models. The iterative process continues until meeting an end-of-sequence token, resulting in a full sequence of response tokens (K is the sequence length of response tokens), and then detokenized to form the natural language response.

Although MLLMs have demonstrated impressive performance on a wide range of multi-modal tasks, they are still prone to hallucination contents [16] without expectation. This phenomenon refers to the model's tendency to generate responses that are inconsistent with the visual input.

IV. Methodology

To address the hallucination challenge in MLLMs, we propose the Complementary Visual Grounding (CVG) framework, as shown in Figure 4. CVG can be understood through the lens of guided information flow, where we aim to maximize the signal from query-relevant visual evidence while actively suppressing two key factors of hallucination: insufficient visual context and progressive textual drift. The following three modules are designed to implement this principle at the input, processing, and output stages of generation, respectively.

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First, the Visual Token Disentanglement module separates visual tokens into two complementary sets to form two distinct evidence branches. Second, the Dynamic Attention Calibration module adaptively adjusts attention weights between visual and textual contents during generation. Finally, the Grounding Flow Fusion module integrates outputs of two branches by contrastive decoding to produce a final prediction.

A. Visual Token Disentanglement

As suggested by our observations in Figure 2, hallucinations often arise from irrelevant or insufficient visual context. The primary goal of this module is to identify and separate visual tokens that are most critical for answering a given query. By disentangling the visual tokens into a set that directly supports the query and a complementary set containing distracting information, we can guide the model toward a more visually-grounded generation. This disentanglement is achieved by computing a hybrid significance score for each visual token.

1) Attention-Aware Token Selection: According to the observation in Figure 2, the attention scores of the MLLMs can serve as a powerful proxy for the importance of each visual token with respect to a given query. The high attention weights between query tokens and visual tokens suggest a strong semantic relationship. To capture the relevance of visual regions to the user's query, we analyze the cross-attention between textual query tokens $\boldsymbol{X}_q$ and visual tokens $\boldsymbol{X}_v$:

$$\boldsymbol{A}_{\text{Q2I}} = \text{softmax}\left(\boldsymbol{Q}_q \boldsymbol{K}_v^\top / \sqrt{d_k}\right) \in \mathbb{R}^{M\times N}, \tag{5}$$

where the $\boldsymbol{Q}_q$ is derived from query tokens $\boldsymbol{X}_q$, $\boldsymbol{K}_v$ is derived from visual tokens $\boldsymbol{X}_v$, M is the number of query tokens, and N is the number of visual tokens.

The last query token $(\boldsymbol{X}_q)_M$ has a special position to influence the generation of the initial responses, and represents the query's overall intent from the attention modules. Therefore, we extract its attention distribution over all visual tokens, averaging across multiple attention heads and intermediate layers [34] to obtain the query-to-image attention scores:

$$\boldsymbol{a}_{\text{cross}} = \frac{1}{|\mathcal{L}|}\sum_{l\in\mathcal{L}} \text{AvgHead}((\boldsymbol{A}^l_{\text{Q2I}})_{M,1:N}) \in \mathbb{R}^N, \tag{6}$$

where $(\boldsymbol{A}^l_{\text{Q2I}})_{M,1:N}$ denotes the attention scores from the M-th query token (last token) to all N visual tokens in the l-th layer, $\mathcal{L}$ represents the set of selected layers, and AvgHead$(\cdot)$ denotes the averaging operation across attention heads.

Besides, relying solely on query-to-image attention is insufficient, as it may overlook intrinsically important regions of the image not explicitly mentioned in the query. To account for this, we incorporate a query-agnostic saliency derived from the self-attention within the vision encoder:

$$\boldsymbol{A}_{\text{I2I}} = \text{softmax}\left(\boldsymbol{Q}_v \boldsymbol{K}_v^\top / \sqrt{d_k}\right) \in \mathbb{R}^{N\times N}, \tag{7}$$

Specifically, we utilize the attention from the *[CLS]* token, which aggregates global visual information from visual tokens. Similar to the cross-attention score, we extract the attention from the intermediate layers of the vision transformer, and compute an average across attention heads and layers:

$$\boldsymbol{a}_{\text{self}} = \frac{1}{|\mathcal{L}|}\sum_{l\in\mathcal{L}} \text{AvgHead}((\boldsymbol{A}^l_{\text{I2I}})_{N,1:N}) \in \mathbb{R}^N, \tag{8}$$

where $(\boldsymbol{A}^l_{\text{I2I}})_{N,1:N}$ represents the attention from the *[CLS]* token to all visual tokens. Finally, we combine these two scores to form a comprehensive significance score vector: $\boldsymbol{a} = \boldsymbol{a}_{\text{cross}} + \boldsymbol{a}_{\text{self}} \in \mathbb{R}^N$. Each element $\boldsymbol{a}_i$ quantifies the overall importance of the i-th visual token.

2) Complementary Evidence Branches: Leveraging the significance score $\boldsymbol{a}$, we can partition the set of visual tokens into two complementary subsets. These subsets form the foundation of our dual-branch approach: one branch built on relevant visual evidence and the other on evidence. This partitioning is governed by a selection ratio hyperparameter $\alpha \in (0, 1)$. Therefore, we can define the two sets:

The Chosen Set ($\boldsymbol{X}_v^{chosen}$): Comprises the top αN visual tokens with the highest significance scores. This branch is designed to contain the most relevant visual evidence, representing a high-fidelity, low-uncertainty information stream.

The Rejected Set ($\boldsymbol{X}_v^{rejected}$): Consists of the remaining $(1-\alpha)N$ tokens with the lowest scores. This branch captures background, irrelevant, or potentially confounding visual content, thus embodying a low-fidelity, high-uncertainty stream.

Critically, we operate the partitioning at the raw image patch level, before the patches are fed into the vision encoder. We create two sets of patches, I_{chosen} and I_{rejected}, which are then independently processed by the vision encoder and projector. This ensures that the resulting visual embeddings for each branch are generated based only on the selected visual context, preventing information leakage between the two branches. The resulting visual token sequences of each branch are:

$$\begin{aligned}\boldsymbol{X}_v^{\text{chosen}} &= \text{ViT}(I_{\text{chosen}}) \in \mathbb{R}^{\alpha N\times d},\\ \boldsymbol{X}_v^{\text{rejected}} &= \text{ViT}(I_{\text{rejected}}) \in \mathbb{R}^{(N-\alpha N)\times d},\end{aligned} \tag{9}$$

In this manner, we construct two complementary sequences of visual tokens, which serve as parallel evidence branches for MLLMs, enabling the model to explicitly reason about what is visually present and relevant versus what is not, a core mechanism for the subsequent hallucination mitigation.

B. Dynamic Attention Calibration

As in the observation in Figure 3, the hallucination is relevant to the response length. When MLLM generates more text, its attention can progressively shift away from the initial image and toward its own previously generated tokens. This over-reliance on the internal language prior, at the expense of visual grounding, is a significant factor of hallucination in tasks requiring detailed descriptions or reasoning.

To counteract this, we propose the Dynamic Attention Calibration module to re-anchor the model's focus onto the visual tokens and reduce the overweighting of textual tokens. This calibration is applied dynamically, with its intensity increasing as the response grows longer. During the generation of the

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i-th response token, it attends to the concatenated sequence of visual tokens, query tokens, and previous response tokens:

$$\boldsymbol{S} = \left(\boldsymbol{Q}_r(\boldsymbol{K}_v + \boldsymbol{K}_q + \boldsymbol{K}_r)^\top / \sqrt{d_k}\right) \in \mathbb{R}^{K\times(N+M+K)}, \tag{10}$$

where attention weights are $\boldsymbol{A} = \text{softmax}(\boldsymbol{S})$. Instead of altering equation weights, we operate on the matrix $\boldsymbol{S}$, which is more stable. We introduce a calibration term δ, which is added to visual tokens and subtracted from the textual tokens. The magnitude is designed to grow with the generation step:

$$\delta_i = \beta \cdot \left(1 - \exp(-i/\tau)\right) \tag{11}$$

where β controls the maximum calibration strength, the latter function ensures a smooth increase (from 0 to 1), and the τ modulates the calibration increasing rate, allowing the effect to be more pronounced in later generation stages where drift is most severe. Crucially, the application of attention calibration is tailored to our two evidence branches:

Chosen Branch: To enforce continuous visual grounding, we calibrate attention towards chosen visual tokens.

$$(\bar{\boldsymbol{S}}^{\text{chosen}})_{ij} = \begin{cases} (\boldsymbol{S}^{\text{chosen}})_{ij} + \delta_i & \text{if token } j \text{ is visual,} \\ (\boldsymbol{S}^{\text{chosen}})_{ij} - \delta_i & \text{if token } j \text{ is textual,} \end{cases} \tag{12}$$

Rejected Branch: Conversely, since this branch is built on weak evidence, we apply an inverse calibration to penalize attention to the rejected visual tokens:

$$(\bar{\boldsymbol{S}}^{\text{rejected}})_{ij} = \begin{cases} (\boldsymbol{S}^{\text{rejected}})_{ij} - \delta_i & \text{if token } j \text{ is visual,} \\ (\boldsymbol{S}^{\text{rejected}})_{ij} + \delta_i & \text{if token } j \text{ is textual,} \end{cases} \tag{13}$$

The dual calibration strategy creates two divergent generation pathways. The chosen branch is steered towards a factually grounded response, while the rejected branch is towards a potentially fabricated one. This starkly contrasting behavior provides a powerful signal for the fusion module to distinguish between faithful and hallucinatory content.

C. Grounding Flow Fusion

Having generated two distinct output distributions from the complementary evidence branches, the final module, Grounding Flow Fusion (GFF), integrates them to produce a single, robustly grounded output. It leverages the grounding knowledge of the chosen branch while using the disturbed output of the rejected branch to actively penalize hallucinations.

1) Plausibility Filtering: As a preliminary step, we constrain the vocabulary to a set of plausible next tokens to improve efficiency and prevent unlikely predictions. Following the previous constraint method [35], we first generate a candidate set $\mathcal{V}_c$, using the chosen branch's output distribution. A token y is considered a candidate if its probability, conditioned on the visual input $\boldsymbol{X}_v$ and query $\boldsymbol{X}_q$, exceeds a certain fraction of the maximum probability in the distribution:

$$\mathcal{V}_c = \left\{ y \mid p(y \mid \boldsymbol{X}_v^{\text{chosen}}, \boldsymbol{X}_q) \geq \eta \cdot \max_{y'} p(y' \mid \boldsymbol{X}_v^{\text{chosen}}, \boldsymbol{X}_q) \right\}, \tag{14}$$

where y and y' are the final and initial output tokens, respectively. And $p(y \mid \boldsymbol{X}_v^{\text{chosen}}, X_q)$ is the probability distribution of the chosen branch, and can be marked as $\boldsymbol{p}_{\text{chosen}}$. Similarly, $\boldsymbol{p}_{\text{rejected}}$ is the distribution of the rejected branch. We set $\eta = 0.2$ to follow previous work [23]. All subsequent decoding operations are confined to this candidate set.

2) Adaptive Contrastive Decoding: We exploit the adaptive contrastive decoding on the complementary branches. It operates on the logit outputs (before softmax normalization) for numerical stability. The goal is to penalize hallucinations only when there is a clear disagreement between two branches.

First, we measure the divergence between the output probability distributions from the chosen and rejected branches, $\boldsymbol{p}_{\text{chosen}}$ and $\boldsymbol{p}_{\text{rejected}}$, by the Kullback-Leibler (KL) divergence. It can be marked as $D_{KL}(\boldsymbol{p}_{\text{chosen}} \| \boldsymbol{p}_{\text{rejected}})$, which tells us how much the rejected branch has diverged from the grounded, chosen branch. Based on this divergence, we adopt the fusion strategy on the logit outputs of the two branches.

Case 1: Divergent Branches: A high divergence indicates that the rejected branch is likely producing hallucinations. In this scenario, we apply contrastive decoding to amplify the signal from the chosen branch while penalizing the output of the rejected branch. Inspired by classifier-free guidance [36], the final logits are computed by steering the chosen logits away from the rejected ones (we set $\gamma = 1$ for simplicity):

$$\begin{aligned} \text{logit}_{\text{final}} = \text{logit}_{\text{chosen}} + \gamma \cdot (\text{logit}_{\text{chosen}} - \text{logit}_{\text{rejected}}) \\ \text{if } D_{KL}(\boldsymbol{p}_{\text{chosen}} \| \boldsymbol{p}_{\text{rejected}}) \geq \epsilon, \end{aligned} \tag{15}$$

Case 2: Convergent Branches: A low divergence suggests that both branches agree on the next token distribution. In this case, the visual evidence is unambiguous, and applying a penalty is unnecessary. Hence, we rely on the chosen branch:

$$\text{logit}_{\text{final}} = \text{logit}_{\text{chosen}} \quad \text{if } D_{KL}(\boldsymbol{p}_{\text{chosen}} \| \boldsymbol{p}_{\text{rejected}}) < \epsilon, \tag{16}$$

Finally, the next token is generated by applying the plausibility mask from $\mathcal{V}_c$ to these final logits and then passing them through a softmax function to obtain the final probability distribution, from which we sample the next token. This adaptive fusion ensures that our model robustly suppresses hallucinations when visual contexts are contested, while preserving generation quality when it is not.

D. Discussion

1) How CVG Mitigates the Hallucination Causes: The CVG framework is designed to correspond to two key causes of hallucinations. First, the Visual Token Disentanglement (VTD) module in Section IV-A addresses the *Insufficient Visual Context* by disentangling text-relevant visual tokens from intermediate layers, where significant semantic information is preserved. Second, the Dynamic Attention Calibration (DAC) module in Section IV-B mitigates the *Progressive Textual Drift* by dynamically calibrating cross-modal attention at each decoding step to prevent attention deviation from visual contents. Finally, the Grounding Flow Fusion (GFF) module in Section IV-C integrates the outputs of VTD and DAC into a unified flow, ensuring that the generation is guided by both enriched visual contexts and calibrated attention.

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2) Differences From Previous Methods: Although some contrastive decoding [23] and attention-rollback [25] methods also aim to mitigate hallucination, they typically rely on pixel-level visual contents and local attention distribution, which limits global semantic reasoning. Our framework instead operates at the semantic-level visual contents with dynamic global attention, enabling stronger generalization. Concretely, (a) We perform token-level disentanglement that separates chosen and rejected visual evidence, which is absent in previous decoding methods. (b) Our calibration mechanism combines attention reweighting with calibrated logit fusion, providing a semantic-aware adjustment rather than the dual-model contrast used in contrastive decoding. (c) Unlike attention rollback, which simply restores earlier attention, ours reconstructs attention distributions based on significant visual contents.

V. Experiments

A. Implementation Details

1) Basic Models: We conduct experiments on the LLaVA [28], a series of powerful and widely-adopted open-source MLLMs. The LLaVA utilizes the ViT from CLIP [40] as the image encoder, and uses the LLaMA-series [41] as the language decoder. (1) LLaVA-1.5 [42]: it is the most popular baseline model, we use the variants built upon the Vicuna-v1.5 (7B & 13B) language models [43]. (2) LLaVA-Next [2] (also called LLaVA-1.6): it splits the original high-resolution images into many sub-images to encode visual tokens (we apply the token disentanglement of Section IV-A for each sub-image).

2) Attention Layers: The layer selection was tailored to the model size: For Visual Token Disentanglement, the attention scores were aggregated from the intermediate layers of the LLM's transformer blocks. Specifically, we used layers #19-#22 for the 7B models and layers #24-#27 for the 13B models. And we choose the #16-#18 layers for the vision transformer in CLIP. For Dynamic Attention Calibration, the calibration was applied to the layers responsible for higher-level semantic integration, spanning from the intermediate to the deep layers (layers #20-#32 for 7B models and #24-#40 for 13B models).

3) Hyper-Parameter: The visual token selection ratio $\alpha = 0.7$, the attention calibration strength $\beta = 0.2$, calibration increasing rate $\tau = 10$, and the threshold of KL-divergence $\epsilon = 0.01$.

B. Comparative Methods

As CVG is a training-free strategy, we compare it with several state-of-the-art methods from the same category in hallucination mitigation. They also leverage the technical idea of contrastive decoding or attention adjustment, providing a strong basis for demonstrating the effectiveness of CVG.

- **Vanilla** serves as our primary baseline, representing the standard output from the original, unmodified model without any specialized decoding strategy.
- **VCD** [23] proposes a visual contrastive decoding by contrasting the output distribution from an original image with that from a distorted version. This technique reduces the model's over-reliance on language bias.
- **OPERA** [24] proposes an attention adjustment that identifies and penalizes "over-trust" patterns during decoding. It corrects the generation path by implementing a rollback and re-selection mechanism for attention weights.
- **PAI** [25] combines contrastive decoding and attention adjustment strategy. It contrasts distributions between standard image-text inputs and text-only inputs while amplifying the attention weights on images.

Existing methods apply contrastive decoding at the pixel level of images or statically adjust attention within local sequences. In contrast, our proposed CVG performs contrastive decoding at the visual semantic level and adaptively calibrates attention across entire sequences. Therefore, CVG could achieve a holistic understanding of visual content, leading to superior performance and generalization ability.

C. Benchmark Evaluation

We evaluate our framework across a diverse set of recent benchmarks designed for MLLMs, categorizing them into two areas: hallucination mitigation and general ability.

1) Hallucination Mitigation Benchmark:

- **POPE** [37] is for evaluating the object hallucination by object-existence questions from three splits (random, popular, adversarial). We compute the average scores of the accuracy metrics for all questions.
- **MME-Hal** [38] serves as a subset of the MME benchmark with yes/no questions beyond the hallucination (object-level and attribute-level), including a measure of object existence, count, position, and color.
- **CHAIR** [26] is designed as the image captioning task. The object mentioned in the caption that does not exist in the ground-truth set is considered a hallucinated object. Its metric comprises two evaluation dimensions: instance-level (CHAIR_I) and sentence-level (CHAIR_S).
- **AMBER** [39] is the LLM-free multi-dimensional benchmark. It can be used to evaluate both generative and discriminative tasks. The benchmark designs four metrics (CHAIR, Cover, Hal, and Cog) on the generative task, and uses the accuracy metric on discriminative tasks.

2) General Ability Benchmark:

- **VQA** contains various task-oriented Visual Question Answer (VQA) benchmarks. VQA-v2 [45] and GQA [46] evaluate the visual perception capabilities on open-ended short answers. VizWiz [47] evaluates on visual questions asked by visually impaired people. ScienceQA [48] evaluates scientific question answering. TextVQA [49] contains text-rich visual question answering.
- **MME** [38] serves as a holistic benchmark with yes/no questions. It comprises ten perception sub-tasks (MME-P) and four cognition ones (MME-C). Performance is quantified via the combined accuracy metric.
- **MM-Bench** [50] evaluates the general perception capabilities. It applies all-around shuffling to multiple-choice answers for the robustness of the answers.
- **LLaVA-Wild** [2] contains multi-modal questions covering diverse contexts (indoor/outdoor scenes, memes, paintings), designed to evaluate MLLMs' ability to handle

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TABLE I

PERFORMANCE COMPARISON ON THREE DISCRIMINATIVE HALLUCINATION BENCHMARKS. (1) RAN., POP., ADV REPRESENTS THE ACCURACY SCORE OF THE RANDOM, POPULAR, ADVERSARIAL SET IN THE POPE BENCHMARK [37], RESPECTIVELY. (2) EXIT., CNT., POS., COL REPRESENTS THE MME ACCURACY SCORE OF THE EXISTENCE, COUNT, POSITION, AND COLOR SET IN THE MME BENCHMARK [38], RESPECTIVELY. (3) ATTR., STAT., REL. REPRESENTS THE ACCURACY SCORE OF THE ATTRIBUTE, STATE, AND RELATION OF THE DISCRIMINATIVE PART IN THE AMBER BENCHMARK [39], RESPECTIVELY.FOLLOWING PREVIOUS WORK [23], ALL RESULTS ARE AVERAGED OVER FIVE RUNS WITH A STANDARD SAMPLING (TEMPERATURE=1.0)

Method	POPE [37]									MME-Hal [38]				AMBER [39]		
	MSCOCO			A-OKVQA			GQA			Object-level		Attribute-level		Discriminative part		
	ran.	*pop.*	*adv.*	*ran.*	*pop.*	*adv.*	*ran.*	*pop.*	*adv.*	*exis.*	*cnt.*	*pos.*	*col.*	*attr.*	*stat.*	*rel.*
LLaVA-1.5-7B [2]																
Vanilla [2]	83.3	81.9	79.0	83.4	79.9	74.0	83.7	78.2	75.1	175.7	124.7	114.0	151.0	71.3	69.6	57.5
VCD [23]	87.7	85.4	80.9	86.2	81.9	75.0	86.7	80.7	76.1	184.7	138.3	128.7	153.0	72.4	70.5	58.1
OPERA [24]	87.5	85.4	80.2	86.3	82.1	74.6	86.4	80.3	75.4	185.0	131.7	128.0	156.0	72.8	70.9	58.4
PAI [25]	88.2	85.8	81.1	86.9	82.5	74.8	87.4	81.2	76.4	170.0	131.0	127.7	155.0	72.6	70.3	58.2
CVG	**89.1**	**86.4**	**82.3**	**88.4**	**83.6**	**77.5**	**88.2**	**82.0**	**77.3**	**190.0**	**141.0**	**131.7**	**168.3**	**73.4**	**71.6**	**59.0**
LLaVA-1.5-13B [2]																
Vanilla [2]	83.3	82.5	80.0	83.9	81.8	76.5	83.6	80.1	77.8	180.0	127.7	114.0	160.0	75.7	74.3	62.6
VCD [23]	87.0	85.1	81.9	86.3	83.8	77.9	86.9	84.1	80.0	185.0	131.7	125.0	170.0	76.8	75.4	65.4
OPERA [24]	86.6	84.8	81.4	86.0	83.5	78.1	86.6	84.2	80.3	185.0	131.0	124.7	165.0	77.1	75.7	66.5
PAI [25]	87.2	85.9	83.1	86.6	84.8	78.5	87.4	84.6	80.7	184.6	138.3	127.7	168.3	77.2	75.3	67.0
CVG	**88.0**	**86.4**	**83.8**	**87.5**	**86.3**	**78.9**	**87.7**	**84.6**	**81.0**	**190.0**	**143.3**	**133.7**	**173.0**	**77.6**	**77.3**	**70.1**
LLaVA-Next-7B [28]																
Vanilla [28]	85.6	84.4	82.4	85.7	83.2	77.4	84.8	79.4	77.1	165.0	115.0	111.7	170.0	74.2	73.0	59.7
VCD [23]	87.2	85.6	83.7	88.7	85.7	80.5	86.5	80.7	78.6	185.0	155.0	128.3	180.0	75.2	73.2	60.3
OPERA [24]	86.8	85.5	83.6	88.1	85.5	80.2	86.0	80.5	78.3	170.0	155.0	126.6	180.0	75.5	73.4	60.6
PAI [25]	87.3	86.1	84.0	88.7	85.9	80.7	86.6	81.0	78.9	190.0	151.7	143.3	185.0	75.8	74.0	60.5
CVG	**88.5**	**87.3**	**85.4**	**90.0**	**87.2**	**81.7**	**87.9**	**82.5**	**80.6**	**185.0**	**158.3**	**153.3**	**185.0**	**75.9**	**74.5**	**61.2**
LLaVA-Next-13B [28]																
Vanilla [28]	86.2	84.8	82.4	87.1	84.4	77.9	86.1	81.6	77.4	185.0	130.0	116.7	178.3	77.2	75.8	62.4
VCD [23]	87.3	86.0	83.3	88.2	85.8	78.8	87.5	83.0	79.8	190.0	160.0	143.3	195.0	77.6	76.7	67.6
OPERA [24]	87.6	85.9	83.1	88.6	86.0	79.0	87.8	83.4	80.9	190.0	158.3	143.3	190.0	78.3	76.4	68.5
PAI [25]	87.9	86.2	83.4	88.0	85.7	78.7	87.2	83.6	80.5	195.0	160.0	151.7	190.0	78.0	77.8	70.3
CVG	**90.4**	**88.7**	**85.4**	**88.6**	**87.3**	**79.5**	**88.3**	**85.1**	**80.8**	**195.0**	**163.3**	**160.0**	**195.0**	**79.3**	**78.6**	**72.8**

challenging tasks and adapt to new domains. Scores are generated by GPT-4 against reference answers.

- **MM-Vet** [51] evaluate the model's capabilities in engaging in visual conversations on a diverse range of tasks, and evaluate the correctness and helpfulness of responses.

Following previous work [2], [23], we employed either nucleus sampling with a temperature of 1.0 or greedy decoding for different benchmarks. The maximum number of new tokens for generation was set to 512 across all experiments.

D. Comparison on Hallucination Mitigation

We evaluate our CVG framework on a suite of hallucination benchmarks in two distinct settings: short-form discriminative tasks and long-form generative tasks.

1) Discriminative Tasks: This evaluation setting involves queries that probe the model's understanding of object existence and attributes, requiring short, factual answers. As detailed in Table I, our CVG framework establishes a new state-of-the-art, consistently outperforming existing methods on prominent benchmarks including POPE [37], MME-Hal [38], and AMBER [39]. Crucially, these performance gains are not confined to a single model but are realized across both LLaVA-1.5 [2] and the more advanced LLaVA-Next [28] backbones, scaling effectively from 7B to 13B parameters. These improvement demonstrates the efficacy of our Visual Token Disentanglement module in correctly grounding the model's responses in salient visual evidence.

TABLE II

PERFORMANCE COMPARISON ON THE TWO GENERATIVE HALLUCINATION BENCHMARKS.THE CHAIR [26] AND AMBER [39] BENCHMARKS ASSESS HALLUCINATION IN OPEN-ENDED IMAGE CAPTIONING TASKS

Method	CHAIR [26]			AMBER [39] (Generative task)			
	$CHAIR_S\downarrow$	$CHAIR_I\downarrow$	Recall↑	CHAIR↓	Cover↑	Hal↓	Cog↓
LLaVA-1.5-7B [2]							
Vanilla [2]	57.0	17.5	73.5	11.2	50.6	47.9	4.2
VCD [23]	50.8	14.6	75.4	10.1	50.9	45.2	4.0
OPERA [24]	46.4	14.2	75.9	10.3	51.0	44.3	4.0
PAI [25]	45.5	14.0	76.0	9.7	51.1	43.6	3.9
CVG	**44.1**	**13.4**	**76.5**	**8.5**	**51.8**	**40.3**	**3.7**
LLaVA-1.5-13B [2]							
Vanilla [2]	53.9	16.3	73.9	10.6	51.0	46.9	4.4
VCD [23]	48.3	15.4	75.1	9.2	51.5	43.6	3.7
OPERA [24]	46.1	14.9	74.8	8.9	51.8	42.9	3.8
PAI [25]	45.8	14.5	75.0	8.6	52.2	40.1	3.6
CVG	**43.2**	**12.6**	**76.2**	**7.1**	**53.4**	**35.1**	**3.2**
LLaVA-Next-7B [28]							
Vanilla [28]	42.6	12.4	62.1	10.8	61.1	56.8	5.1
VCD [23]	40.3	11.3	63.2	9.4	62.3	53.2	4.7
OPERA [24]	39.2	10.7	63.7	8.9	62.6	52.6	4.6
PAI [25]	39.9	11.1	62.4	8.6	63.3	51.5	4.5
CVG	**38.6**	**10.3**	**64.4**	**7.8**	**65.8**	**48.4**	**4.2**
LLaVA-Next-13B [28]							
Vanilla [28]	41.2	11.1	63.4	11.3	60.6	55.9	4.9
VCD [23]	40.2	10.2	63.5	9.7	61.5	51.2	4.3
OPERA [24]	40.4	10.7	62.6	10.2	60.9	52.3	4.4
PAI [25]	39.8	10.0	63.9	8.6	62.4	50.8	4.1
CVG	**38.6**	**9.6**	**64.2**	**7.9**	**62.1**	**50.4**	**3.7**

2) Generative Tasks: This more challenging setting assesses the model's ability to generate detailed image descriptions without fabricating content, a common issue in long-sequence generation. Table II shows that CVG achieves superior performance on CHAIR [26] and AMBER [39] benchmarks. This highlights the critical role of our Dynamic

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TABLE III
PERFORMANCE COMPARISON ON THE TWO HALLUCINATION BENCHMARKS WITH THE MODEL-BASED EVALUATION METRICS

Method	MM-Hal [22]		HallusionBench [22]		
	Score ↑	Hal-Rate ↓	Figure Acc ↑	Easy Acc ↑	Hard Acc ↑
LLaVA-1.5-13B [2]					
Vanilla [2]	2.38	0.55	24.8	49.6	29.7
VCD [23]	2.31	0.54	25.1	49.9	29.5
CVG	**2.48**	**0.53**	**25.6**	**50.3**	**30.7**
Qwen2-VL-2B [44]					
Vanilla [2]	2.17	0.59	15.3	36.4	17.6
VCD [23]	2.23	0.59	15.8	37.5	18.6
CVG	**2.26**	**0.57**	**16.5**	**37.9**	**18.9**
Qwen2.5-VL-3B [29]					
Vanilla [2]	2.62	0.49	16.5	38.0	16.1
VCD [23]	2.73	0.47	17.9	40.4	16.0
CVG	**3.14**	**0.42**	**18.8**	**41.2**	**16.8**
Qwen2.5-VL-7B [29]					
Vanilla [2]	3.35	0.41	22.5	50.8	25.6
VCD [23]	3.42	0.40	22.4	49.7	25.4
CVG	**3.55**	**0.38**	**24.0**	**51.4**	**26.3**

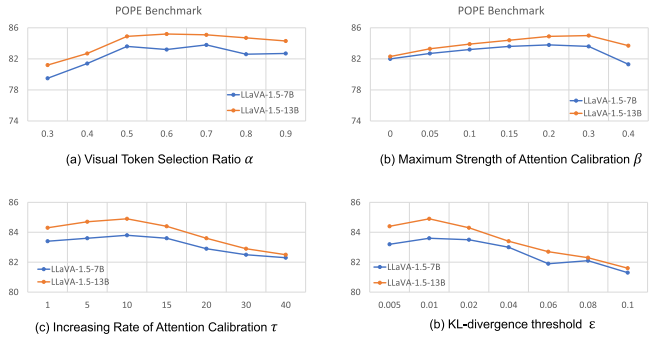


Fig. 5. The impact of four significant hyperparameters in our framework. (a) The visual token selection ratio α between chosen and rejected tokens in Eq. 9. (b) The maximum strength β for attention calibration in Eq. 11. (c) The increasing rate for attention calibration τ in Eq. 11. (d) The KL-divergence threshold ϵ for contrastive decoding in Eq. 15.

Attention Calibration module, which prevents the model from exhibiting semantic drift during prolonged generation. By continually re-anchoring the model's focus on the visual context, our framework effectively mitigates the tendency to hallucinations.

E. Comparison on General Ability

A critical test for any hallucination mitigation technique is whether it preserves or enhances the model's general capabilities, avoiding the common performance trade-offs. We demonstrate that CVG not only prevents this degradation but actually improves performance on task-oriented VQA and complex instruction-following benchmarks.

1) Task-Oriented VQA: As shown in Table IV, applying CVG boosts performance on several traditional VQA benchmarks compared to both the base models and previous hallucination-focused works. This indicates that better visual grounding directly contributes to more accurate question-answering.

While our CVG gets great improvements, we acknowledge the performance degradation on some OCR tasks, such as TextVQA [49]. An explanation is that the visual token disentanglement module may incorrectly classify patches containing fine-grained textual patterns as "rejected", thus discarding essential character-level information. In addition, text regions typically consist of high-frequency and fine-grained details, and the subsequent CVG inevitably introduces the semantic aggregation process that further weakens these details, making texts harder to distinguish. Despite this limitation in text-heavy scenarios, the overall trend demonstrates that CVG substantially achieves great gains in various VQA tasks.

2) Instruction-Following: We further evaluated our method on modern benchmarks like MM-Bench [50] and MM-Vet [51], which are designed to assess complex reasoning and instruction-following in MLLMs. Table IV reveals that CVG brings significant improvements to the base models in these challenging scenarios. This is particularly noteworthy as previous hallucination mitigation methods often struggle or fail to show benefits on such complex evaluations. The success of CVG underscores its strong generalization ability, proving that it is not a narrow fix but a foundational improvement that enhances model reliability across a wide spectrum of tasks.

F. Ablation Study

To comprehensively analyze each component of our CVG framework, we conduct a series of ablation experiments. Unless otherwise specified, all ablations are performed using the LLaVA-1.5-7B model [28] as the baseline.

1) Impact of Sampling Strategies: We first analyze the robustness of our framework by applying it across various sampling strategies. We test five common sampling approaches: greedy sampling, beam search (k = 5), sampling with a lower temperature (t = 0.5), Top-P (nucleus) sampling (p = 0.8), and Top-K sampling (k = 20). The Table V demonstrates that CVG yields consistent performance gains in hallucination mitigation regardless of the decoding strategy employed. This proves that the benefits are fundamental despite the specific sampling.

2) Impact of Visual Token Disentanglement: The Visual Token Disentanglement module hinges on an effective strategy for identifying significant visual tokens. We have tried different strategies to select visual tokens, as shown in Table VI. The results clearly indicate that combining both attention sources yields the most significant performance gains, confirming that both intrinsic visual saliency and query-specific relevance are crucial for creating effective evidence branches. Besides, although encoding the entire image and partitioning visual patches after the visual encoder yields slight gains, we opted not to adopt this strategy due to the extra computational cost involved (the efficiency discussion is in Section V-G.3).

3) Impact of Attention Layers: As shown in Table VI and Figure 8, we find that using attention from the shallow layers or deep layers is sub-optimal. We believe that the shallow layers primarily capture low-level features, and deep layers can be overly sparse for relevant semantics. The intermediate layers keep the balance between information abstraction and local details [53], resulting in the best performance.

4) Impact of Visual Token Ratio: The hyperparameter of visual token selection ratio α determines the balance between the chosen and rejected evidence branches. We analyze its impact in Figure 5(a). The results show that our framework's

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TABLE IV

PERFORMANCE COMPARISON ON GENERAL CAPABILITY BENCHMARKS FOR MLLMS, INCLUDING TASK-ORIENTED VQA AND INSTRUCTION-FOLLOWING.FOLLOWING PREVIOUS WORK [42], WE USE THE APPROXIMATE GREEDY SAMPLING (TOP-P=0.01, TEMPERATURE=0.01)

Method	Task-oriented VQA benchmark					Instruction-Following benchmark				
	VQA-v2 [45]	G-QA [46]	Vis-Wiz [47]	Science-QA [48]	Text-VQA [49]	MME-P [38]	MME-C [38]	MM-Bench [50]	LLaVa-Wild [2]	MM-Vet [51]
LLaVA-1.5-7B [2]										
Vanilla [2]	78.5	62.0	50.0	66.8	**58.2**	1510	295	**64.3**	65.4	31.1
VCD [23]	78.1	62.3	50.2	65.4	57.5	1473	276	63.8	65.8	30.5
OPERA [24]	77.6	61.4	48.2	64.6	56.1	1420	255	63.2	64.6	29.6
PAI [25]	78.3	62.1	50.6	66.9	57.2	1442	298	64.4	65.3	30.2
CVG	**78.9**	**62.8**	**51.1**	**67.6**	57.4	**1520**	**304**	63.9	**67.1**	**32.0**
LLaVA-1.5-13B [2]										
Vanilla [2]	80.0	63.3	**53.6**	71.6	**61.3**	1531	308	67.7	72.5	36.1
VCD [23]	78.3	63.8	52.9	69.2	59.5	1480	293	65.2	70.4	35.2
OPERA [24]	78.1	62.7	51.8	67.3	58.0	1475	267	63.5	68.8	34.3
PAI [25]	79.2	62.6	53.4	71.2	60.5	1495	286	66.8	72.1	36.0
CVG	**80.5**	**63.9**	53.2	**72.8**	60.1	1522	**326**	**68.2**	**74.7**	**37.2**
LLaVA-Next-7B [28]										
Vanilla [28]	81.8	64.2	**57.6**	70.1	**64.9**	1519	332	67.4	**81.6**	43.9
VCD [23]	80.4	63.2	54.9	68.3	60.4	1525	335	66.2	80.4	42.0
OPERA [24]	79.2	63.1	54.3	68.5	58.9	1445	304	65.1	77.2	41.8
PAI [25]	80.6	63.5	55.6	70.0	61.6	1532	346	67.3	81.5	44.7
CVG	**82.3**	**64.4**	56.8	**71.2**	62.3	**1540**	**355**	**67.9**	80.3	**45.1**
LLaVA-Next-13B [28]										
Vanilla [28]	82.8	65.4	60.5	73.6	**67.1**	1575	326	70.0	87.3	**48.4**
VCD [23]	80.2	64.0	58.6	71.1	65.7	1532	334	68.2	85.5	47.7
OPERA [24]	80.4	63.7	58.5	69.8	62.6	1520	307	67.5	83.8	47.6
PAI [25]	81.5	63.3	59.2	72.6	66.4	1583	349	67.6	83.2	47.8
CVG	**83.3**	**65.9**	**60.6**	**74.5**	66.5	**1590**	**373**	**70.7**	**88.1**	48.2

TABLE V

THE COMPREHENSIVE COMPARISON OF DIFFERENT SAMPLING STRATEGIES.WE USE THE LLAVA-1.5-7B [2] AS THE BASIC MODEL

Method	CHAIR [26]		AMBER-Gen [39]	
	$CHAIR_S\downarrow$	$CHAIR_I\downarrow$	CHAIR↓	Hal↓
Greedy sampling				
Vanilla	52.4	14.9	10.4	44.3
CVG	**40.2**	**11.7**	**7.7**	**36.8**
Beam searching, k=5				
Vanilla	55.5	16.6	10.8	47.2
CVG	**48.2**	**12.3**	**8.4**	**40.0**
Temperature, t=0.5				
Vanilla	56.3	17.4	10.6	46.7
CVG	**43.7**	**13.5**	**8.3**	**40.6**
Top-P, p=0.8				
Vanilla	56.7	17.5	10.6	46.0
CVG	**42.4**	**12.8**	**8.1**	**39.2**
Top-K, k=20				
Vanilla	55.0	16.5	10.3	45.4
CVG	**42.6**	**12.9**	**8.5**	**38.9**

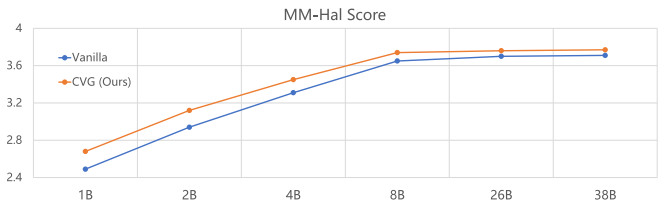


Fig. 6. The scaling-law analysis across different model sizes on the MM-Hal benchmark [22]. We employ the InternVL-2.5 series [52] as the base models, spanning parameter sizes from 1B to 38B.

performance is robust within a moderate range. And performance degrades if it is too high (insufficient negative evidence for contrast) or too low (insufficient positive evidence for grounding). This confirms the importance of maintaining two meaningful, complementary branches.

5) Impact of Dynamic Attention Calibration: As shown in Table VI, removing the attention calibration entirely or using only a fixed calibration yields inferior results compared to our dynamic approach. And the choices of attention layers are important to the calibration operations. We found that the shallow layers are unsuitable, and the best choices are the combination of the middle and deep layers. We believe that shallow layers can't capture abstract semantics [34], and the high-level perception relies on deeper layers.

6) Impact of Calibration Strength and Rate: Figure 5(b) shows that increasing the maximum strength of attention calibration of β from zero provides a clear benefit. But an excessively large β disrupts the model's inherent language capabilities, leading to degraded performance. Figure 5(c) shows that the increasing rate τ should not be too large. Table VI shows that the linear function is competitive with the exponential function in Eq. (11).

TABLE VI

THE IMPACT OF THE THREE CORE MODULES OF OUR CVG FRAMEWORK:THE PERCENTAGE IN PARENTHESES REPRESENTS RELATIVE GAIN.IT MEASURES GAIN RATES FROM VANILLA (0%) TO CVG (100%) UNDER THIS SETTING.LLAVA-1.5-7B HAS 32 TRANSFORMER BLOCK LAYERS

Ablation Settings	CHAIR [26]	
	$CHAIR_S\downarrow$	$CHAIR_I\downarrow$
Vanilla	57.0 +0%	17.5 +0%
Section IV-A: Visual Token Disentanglement Module		
Only self-visual attentions	53.5 +27%	17.0 +12%
Only query-to-image attentions	46.9 +78%	15.2 +56%
Partition visual patches after visual encoder	43.8 +102%	13.1 +107%
Use attention in deep layers (#30-#32)	49.5 +58%	15.4 +51%
Use attention in shallow layers (#8-#10)	54.3 +21%	16.6 +22%
Section IV-B: Dynamic Attention Calibration Module		
Without the attention calibration	48.3 +67%	14.9 +63%
Use the fixed calibration in Eq. (12)(13)	47.6 +73%	15.7 +44%
Use the linear function in Eq. (11)	45.2 +95%	13.8 +90%
Use calibration only in deep layers (#27-#32)	50.0 +54%	14.4 +76%
Use calibration only in shallow layers (#3-#8)	52.3 +36%	15.6 +46%
Section IV-C: Grounding Flow Fusion Module		
Only chosen branch	52.0 +39%	16.4 +27%
Original + chosen branches	55.1 +15%	17.2 +7%
Original - rejected branches	47.5 +74%	15.3 +54%
Original + chosen - rejected branches	43.6 +104%	13.2 +105%
CVG	44.1 +100%	13.4 +100%

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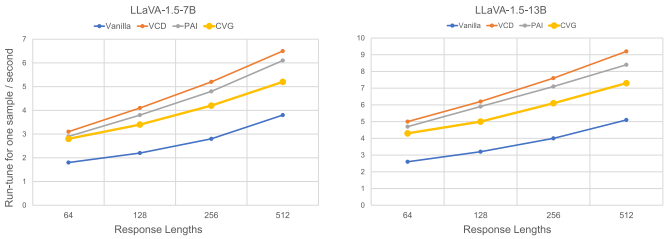


Fig. 7. The comparison of run-time cost with different basic models and response lengths. All experiments are conducted on a single A100 GPU.

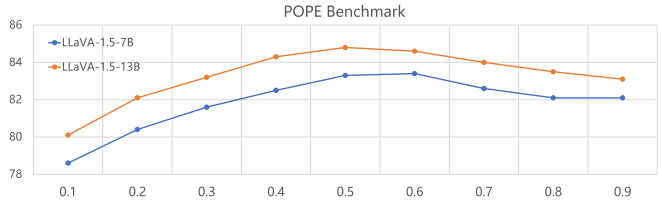


Fig. 8. The performance comparison with different layer selections in the Visual Token Disentanglement module. Given that models vary in depth, we normalize layer indices to the range [0, 1] to represent relative depth. For any specified target layer, we employ a five-layer window to aggregate attention.

7) Impact of Grounding Flow Fusion: The final step of CVG is the fusion of distributions from the two evidence branches. We ablate our Grounding Flow Fusion (GFF) module to assess its effectiveness against alternative strategies, with results in Table VI. The results demonstrate that while simply using the chosen tokens offers some improvement over the vanilla baseline, it is significantly less effective than our contrastive method. Interestingly, incorporating a third branch with the original input does not yield significant further gains but substantially increases computational overhead during inference. Therefore, our proposed dual-branch adaptive contrastive decoding strikes the best balance between performance and efficiency, effectively using the rejected branch as a negative guide to refine the output from the chosen branch.

Besides, the Figure 5(d) shows that the KL-divergence threshold ϵ is important, which represents the critical edge where the "rejected" branch may generate a non-hallucinatory response. Too large a threshold can affect the model's judgment for two complementary branches.

G. Robustness Analysis

To further substantiate the robustness of our framework, we conduct additional analysis. All experiments maintain the same settings as the ablation study unless otherwise noted.

1) More Hallucination Benchmarks: Although Table I, II have conducted experiments on various hallucination benchmarks (e.g., POPE, CHAIR, AMBER), these benchmarks are mostly based on rule-based evaluation metrics. Therefore, we will evaluate on the more comprehensive benchmarks, MM-Hal [22] and HallusionBench [55], which contain model-based evaluation metrics. Table III shows that our CVG gets the best performance compared with the previous works.

2) More Basic Models: We apply CVG to more basic models of MLLMs beyond the LLaVA series. These include the state-of-the-art, Qwen2-VL [44], and Qwen2.5-VL [29] models, which feature distinct architectures and native high-resolution capabilities. As shown in Table III, CVG consistently enhances performance across these varied models. This result demonstrates that our approach is not tailored to a specific architecture but effectively addresses a fundamental weakness in MLLMs. Besides, we also explore the scaling law of InternVL-2.5 [52] in Figure 6. It shows our framework gets consistent improvement as the model sizes grow.

3) Inference Efficiency: A practical consideration for decoding methods is their computational overhead. We evaluate the inference time of CVG against competing methods across different model sizes and response lengths. As shown in Figure 7, our approach introduces a manageable computational overhead that is competitive with other training-free methods. Besides, we conduct the complexity analysis by reporting additional metrics such as FLOPs in Table IX, which shows our framework has a lower computational cost.

Since we run two complementary branches, each branch processes only half of the visual tokens. And the initial forward pass to calculate attention scores in Section IV-A only needs one time. This reduction in the sequence length for the costly attention computations effectively offsets the overhead of the dual-branch structure, making CVG a computationally viable solution without imposing a significant burden.

TABLE VII

IMPACT OF MASKING DIFFERENT VISUAL TOKEN SUBSETS ON HALLUCINATION RATES.THIS EXPERIMENT VALIDATES THAT HALLUCINATION IS TIED TO SALIENT VISUAL EVIDENCE

Masking Tokens Choices	CHAIR [26]	
	$CHAIR_S\downarrow$	$CHAIR_I\downarrow$
Full tokens	57.0	17.5
Random tokens masking, 20%	58.4 +1.4	18.3 +1.0
Random tokens masking, 40%	60.3 +3.3	19.7 +2.2
Chosen tokens masking, 20%	62.5 +5.5	25.3 +7.8
Chosen tokens masking, 50%	75.8 +18.8	35.2 +17.1
Rejected tokens masking, 20%	56.1 -0.9	17.0 -0.5
Rejected tokens masking, 40%	55.6 -1.4	16.9 -0.6

TABLE VIII

ANALYSIS OF HALLUCINATION RATES AS A FUNCTION OF MAXIMUM GENERATION LENGTH. THE RESULTS DEMONSTRATE A CLEAR TREND WHERE HALLUCINATION INCREASES WITH LONGER RESPONSES

Max Tokens	CHAIR [26]		AMBER-Gen [39]	
	$CHAIR_S\downarrow$	$CHAIR_I\downarrow$	CHAIR↓	Hal↓
64	22.3	9.2	5.2	30.4
128	44.5 +22.2	11.3 +2.1	7.8 +2.6	35.6 +5.2
256	52.9 +30.6	15.2 +6.0	9.2 +4.0	40.7 +10.3
512	57.0 +34.7	17.5 +8.3	10.4 +5.2	44.3 +13.9

TABLE IX

COMPARISON OF COMPUTATIONAL COMPLEXITY IN FLOPS (FLOATING POINT OPERATIONS) METRICS.ALL MEASUREMENTS ARE COMPUTED USING CALFLOPS [54] WITH A FIXED GENERATION LENGTH OF 100 STEPS

Methods	TFLOPS ↓	
	LLaVA-1.5-7B	LLaVA-1.5-13B
Vanilla [2]	10.2	19.4
VCD [23]	20.4 (+100%)	38.9 (+100%)
CVG (Ours)	16.7 (+64%)	31.8 (+64%)

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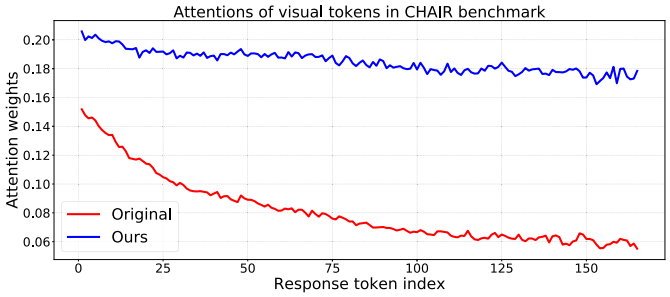


Fig. 9. The curves of the average attention scores of visual tokens at different response generation steps. While the vanilla model's visual attention decays sharply, our model maintains a consistently higher level of attention.

4) Language Prior in Visual Contexts: Our framework is motivated by the hypothesis that hallucinations are triggered by a lack of grounding in salient visual evidence. We analyze the impact of selectively masking different subsets of visual tokens on the base model's performance. As detailed in Table VII, randomly dropping tokens leads to a moderate increase in hallucination. More revealingly, masking the "rejected" tokens (those deemed irrelevant by our selection criteria) has a negligible, sometimes even positive, effect on performance. In stark contrast, masking the "chosen" tokens—the most salient visual evidence—causes a dramatic surge in hallucination across all metrics. This result strongly supports our core premise that the presence and proper utilization of key visual evidence are critical for preventing hallucinations.

5) Language Prior in Textual Length: Next, we examine the relationship between response length and hallucination rates. Table VIII shows a clear and monotonic trend: as the maximum generation length increases, hallucination metrics consistently worsen. Shorter responses remain relatively factual, but as the model generates more text, it increasingly drifts from the visual input, relying more on its internal language prior. This progressive degradation underscores the necessity of our Dynamic Attention Calibration module, which is specifically designed to counteract this effect by continually re-anchoring the model's focus on the visual context.

H. Visualization & Case Study

To provide deeper insight into our CVG framework, we present a series of visualizations and case studies that illustrate the behavior of its core modules.

1) Visualization of Visual Token Disentanglement: To illustrate the effectiveness of our visual token disentanglement module, we visualize the chosen and rejected tokens. We set the selection ratio $\alpha = 0.5$ to maintain the same number of chosen tokens and rejected tokens. Figure 10 shows that for a given query, CVG successfully identifies semantically relevant objects/regions as chosen tokens, while assigning irrelevant areas to the rejected set. Furthermore, Figure 11 demonstrates that the chosen tokens dynamically align with the specific regions of different texutal queries.

2) Visualization of Visual Attention Calibration: We visualize the impact of our Dynamic Attention Calibration module

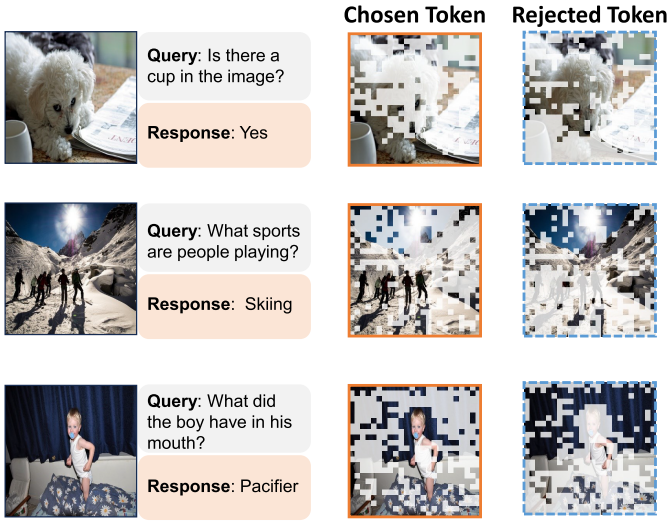


Fig. 10. The visualization of the visual token disentanglement. Given a query, CVG effectively splits image patches into a chosen set containing the key evidence and a rejected set containing trivial ones (the selection ratio $\alpha = 0.5$).

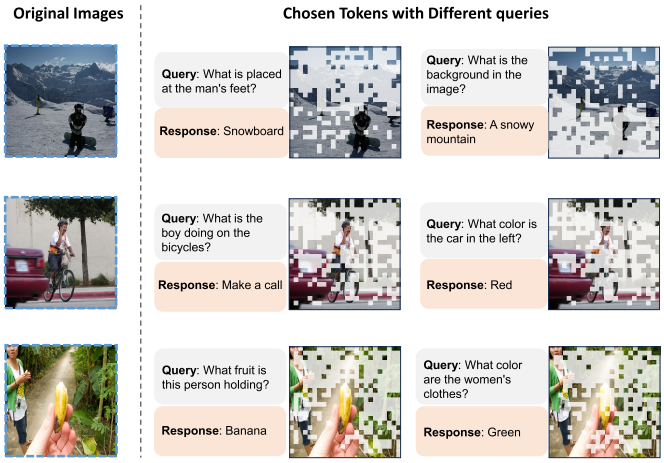


Fig. 11. The visualization of visual token disentanglement with different textual queries. The visual chosen tokens dynamically change based on the query semantics, and focus on the query-specific visual evidence.

on both visual attention scores during generation. Figure 9 plots the aggregated attention paid to visual tokens at each generation step. In the vanilla model, visual attention rapidly diminishes as the response lengthens, indicating a "semantic drift" where the model relies more on generated texts than the image. CVG successfully counteracts this trend, maintaining a more stable level of visual grounding.

3) Visualization of Textual Attention Calibration: Concurrently, Figure 13 shows the intended suppressive effect on textual tokens. CVG reduces the model's over-reliance on its previously generated context by lowering the attention scores directed at textual tokens. This dual-action mechanism—boosting visual attention while suppressing textual over-reliance—effectively re-anchors the model to the visual input throughout the generation process.

4) Cases of Generative Description: Figure 12 presents examples from the CHAIR image captioning benchmark [26].

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Fig. 12. Qualitative examples on the CHAIR benchmark [26]. We highlight hallucinated contents in red, and the correct statements in green. Our framework could effectively eliminate the object and attribute hallucinations.

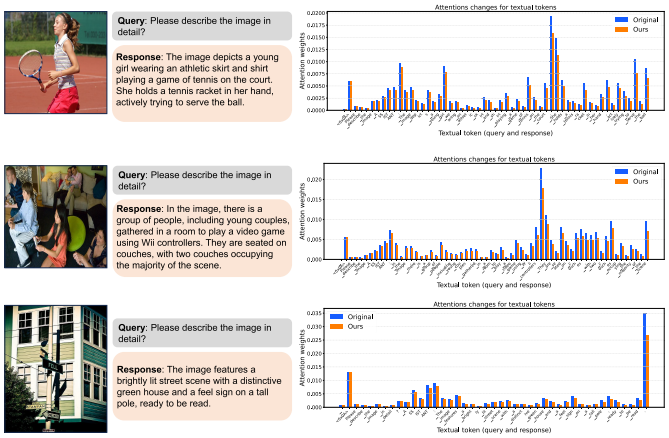


Fig. 13. The comparison of attention scores for each textual token (both query and response). CVG can reduce attention to text, particularly for later tokens, preventing the model from over-relying on its own generated context.

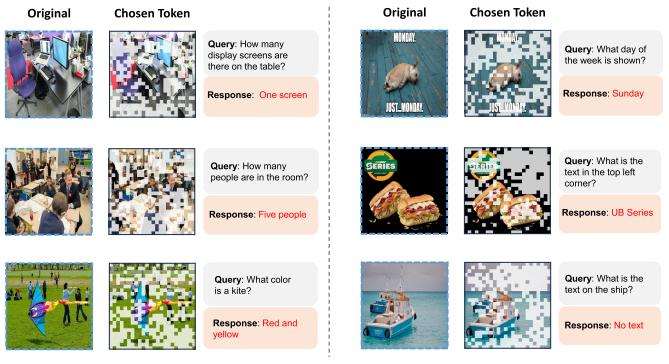


Fig. 14. Visualization of failure cases. In scenarios featuring dense object distribution or fine-grained text in images, our framework may fail to attend to the significant visual tokens, leading to poor responses.

In these cases, the baseline model hallucinates objects or misidentifies attributes. By effectively grounding the generation in the chosen visual evidence, our CVG framework corrects these errors, producing more accurate and faithful descriptions that align with the visual content.

5) Failure Cases: Although our CVG achieves great improvements in most cases, certain limitations remain. As illustrated in Figure 14, it faces challenges when handling dense objects, crowded scenes, or text-rich images. In such scenarios, the attention-based visual token selection mechanism may fail to prioritize fine-grained yet significant visual patches. This can result in the omission of key visual details, leading to occasional hallucinations or inaccurate responses.

VI. Conclusion

In this paper, we focus on the critical hallucination challenge in Multimodal Large Language Models (MLLMs). We identify and empirically validate two key factors: Insufficient Visual Context, where incomplete visual information prompts hallucination, and Progressive Textual Drift, where attention deviates from visual grounding during lengthy generation. To address these issues, we propose a training-free Complementary Visual Grounding (CVG) framework by leveraging the model's intrinsic architecture. Through its core modules for visual token disentanglement, dynamic attention calibration, and grounding fusion, CVG ensures that model outputs remain anchored to visual evidence. Extensive experiments demonstrated that CVG achieves state-of-the-art performance in mitigating hallucinations and improving performance on general benchmarks, confirming its effectiveness and generalizability.

VII. Limitations and Future Work

Although our proposed CVG framework demonstrates strong capability across multiple multi-modal benchmarks, several limitations remain to be addressed in future research. First, the framework exhibits limited performance when handling text-rich images, since the current architecture is not explicitly designed to be text-aware, and its ability in fine-grained textual details is restricted. In future work, introduce a text-aware module without an extra training process, such as using an image-text heatmap, to improve performance on these challenging cases. Furthermore, the framework requires the token selection on visual attention, which relies on a heuristic strategy across different query tokens and transformer layers. It may not be optimal for all images or complex tasks. Future work may explore adaptive attention mechanisms, enabling the framework to dynamically select informative tokens.

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